LifeSync

Bridging the generalization gap in autism skills education

TSA High School Software Development · Tier 1
Team ID: _____ · November 2026
github.com/Nisith-ctrl/software-development-2026

PROBLEM

Learners on the autism spectrum often master a life skill in a structured classroom or clinic, then can't carry it out in the real place: a loud supermarket, a crowded bus, their own bathroom. This is the **generalization gap**. Learning apps tend to widen it. They teach with cartoons that look nothing like the learner's world, they mark mistakes in ways that raise anxiety, they ignore sensory overload, and they stop at "lesson complete" without checking whether the skill works in real life.

CONNECTION TO THE THEME

> *"Develop a software application that improves how people learn, teach, practice, or develop new skills… an engaging, effective, and user-centered software solution."*

- **Learn and practice:** learners rehearse routines step by step inside photos and videos of *their own* places, with errorless prompting so practice never ends in failure.
- **Teach:** teachers, therapists and parents build routines without code and see exactly which step each learner gets stuck on.
- **Develop new skills:** help fades as the learner improves, and real-world check-ins confirm the skill transferred. That transfer is the real goal of practising.
- **User-centred:** tap-only input, calm colours, read-aloud, rewards built on each learner's interests, and a calm-down mode triggered by heart rate.

What's original: LifeSync *measures* the generalization gap. For each routine it compares in-app mastery with logged real-world tries and flags "mastered in the app, not yet in real life", so the team knows where to practise next.

FEATURE LIST

Feature	What it does	Status
"My World" routine builder	Steps with photos or videos of the learner's real environment, tap choices with one right answer, a First/Then reward, and plain-language validation.	**Built**
Errorless learning player	Four prompt levels (Full, Highlight, Delayed, Independent). An off-target tap fades and the right choice is shown, never an error. First/Then board, Now/Next strip, read-aloud.	**Built**
Personal reinforcement	Token board, special-interest animation or soft chime, chosen per learner.	**Built**
Bio-responsive calm-down	A Bluetooth heart-rate device (or simulator) is watched during practice. Sustained high heart rate pauses the routine and opens a dark calm screen: breathing pacer, deep-pressure visual, calm music.	**Built**
Adaptive prompting	Help fades after two clean completions and returns after repeated redirects or a calm break.	**Built**
Real-world check-ins	Grown-ups log real tries. Each routine is labelled Practising, Generalization gap, Transferring or Generalized.	**Built**
IEP dashboard	Sticking points, help level per step, help-over-time chart, CSV export, print view.	**Built**
Photo hotspots	"Tap the toothbrush in the picture" in place of buttons.	**Tier 2**
In-place practice mode	Phone mode that runs the routine during the real task.	**Tier 2**
Learned prompting model	Replaces the fixed fading rules using session data.	**Later**
Accounts and sync	Parent, teacher and therapist roles across devices; watch companion app.	**Later**

DESIGN

- Browser app (React + TypeScript), no install, works offline; data stays on the device (IndexedDB).
- Layered code: a pure **domain** layer holds every rule (session state machine, prompt fading, calm-down trigger, generalization metric); UI and browser services sit on top.
- Grounded in established methods: video modelling, errorless learning, most-to-least prompting.

DEVELOPMENT PRACTICES

- **Requirements:** numbered requirements, each traced to code and tests (docs/REQUIREMENTS.md).
- **Design:** architecture, data model and state diagrams (docs/DESIGN.md).
- **Testing:** 62 automated unit and integration tests, 93% domain coverage, manual test plan; CI runs type check, lint, tests and build on every push.

Run it: `npm install` then `npm run dev`. Demo data is included, so every feature can be shown right away.